dbGap Task 1: Semantic Variable Normalization and Ontological Alignment

Task 1 participants are tasked with developing an AI system that maps raw, unstructured dbGaP study variables to standardized ontology concepts in preferred source vocabularies LOINC, RxNorm, and SNOMED CT, thereby enabling precise, interoperable study discovery and cohort evaluation and other downstream use cases and applications. If no mapping is found in these sources, the Unified Medical Language System® (UMLS®) Metathesaurus, should be used.

The core objective is to ingest raw, unstructured, publicly available dbGaP study data, metadata, data dictionaries, and related public study information, and algorithmically map study variables to standardized reference concepts most appropriate in the context of the study. To support downstream interoperability, mappings should resolve to the established, high-quality reference vocabularies listed above. Systems should select the most appropriate target vocabulary concepts or combination of concepts based on the underlying data type and context, with the goal of maximizing semantic precision, consistency, and classification performance.

To succeed, systems must demonstrate a level of mapping accuracy, contextual understanding, and semantic disambiguation that approaches expert human clinical terminology and ontology-mapping performance.

The following provides an example mapping from dbGaP study variable to its corresponding expected target node in one of the task designated source vocabularies for this challenge task.

- Here, OCEA37L is a variable occurring in the StudyAccession phs000810.v2.p2 (Hispanic Community Health Study /Study of Latinos (HCHS/SOL)). It has a corresponding Variable Description *“Occ Exp (Long Occ) - tobacco smoke (OCEA37L)”*.
    - The following provides a table of the information about this variable, available via the variables page associated with this study; for more details, see https://ncbi.nlm.nih.gov/projects/gap/cgi-bin/GetListOfAllObjects.cgi?study_id=phs000810.v2.p2&object_type=variable .

Study Accession	Variable Accession	Variable Name	Variable Description
phs000810.v2.p2	phv00527205.v1.p2	OCEA37L	Occ Exp (Long Occ) - tobacco smoke (OCEA37L)

The task at hand here is to identify the source-vocabulary concept(s) and code(s) that most accurately represent the meaning of this dbGaP variable in its exact or partial / related form. If no exact or closely related target concept exists in these source vocabularies, then the ONT_NONE should be used as the target.

A dbGaP study variable may fall into one of four mapping categories:

1. **Exact mapping:** The variable maps directly to an existing source-vocabulary concept that captures its full meaning.
2. **Related/Partial mapping:** No direct mapping exists in this case. However, the variable maps directly to an existing source-vocabulary concept that captures the main aspect(s), but not all, of its meaning.
3. **Related Compound:** No direct mapping exists, but the study variable can be broken into multiple distinct concepts that capture components of the variable's meaning. As a result, the variable is decomposed into multiple concepts, with each constituent concept mapped separately, and when combined together capture the entire meaning of the study variable.
    1. **This category includes Partial Related Compound** where only a subset of the concepts included in the variable can be adequately represented by existing source-vocabulary concepts.
4. **No adequate mapping:** No appropriate closely related source-vocabulary concept, or combination of concepts, exists to adequately represent the meaning of the variable. In this case, **ONT_NONE** should be returned.

Expected Response for OCEA37L:

For the example variable OCEA37L given above, the expected response should identify the most appropriate source vocabulary target concept(s). This corresponds to SNOMED CT: 16090771000119104.

- SNOMED CT: 16090771000119104 provides the best semantic match for the OCEA37L variable and is considered a high-confidence **exact mapping**.
- LOINC: 20222-11: this can be considered a related match; however, it is considered a less precise mapping because it does not fully capture the meaning and context of the dbGaP variable (exposure at the primary occupation site).
- In this situation, the correct response is SNOMED CT: 16090771000119104 as it is the best fit. LOINC: 20222-11 is incorrect.

This example illustrates that the objective is **not simply to identify a vocabulary concept containing matching terminology**. Rather, systems should identify the concept that most accurately represents the **semantic meaning of the source variable**, taking into account its description, study context, and related information available in the Challenge Dataset.

Challenge Dataset

Participants will be provided with a fixed snapshot of dbGaP documents, associated metadata, and data dictionaries, including relevant version information where applicable. Responses to Task 1 variable mappings must be limited to and based on this snapshot. The snapshot is derived from official NCBI public repositories and includes the document content and associated metadata required to support the challenge tasks. Participants will therefore work from a common, fixed corpus, rather than independently downloading or accessing documents from dbGaP. The challenge dataset does not require access to controlled-access individual-level data. ***Use of controlled-access individual-level data is strictly prohibited. Any submission found to have used controlled-access individual-level data, either directly or indirectly through an external data source, service, or system, will be disqualified from the challenge.***

The data for this challenge can be downloaded via the following link:

- https://ftp.ncbi.nlm.nih.gov/dbgap/NLM_Challenge/dbgap.tar.gz

This will result in one compressed TAR file being download, dbgap.tar.gz. Participants should unpack the archive to create a base directory named dbgap. The directory contains a README.txt file introducing the snapshot of dbGaP (the database of Genotypes and Phenotypes) studies and their associated objects, together with supporting reference material.

The dataset is organized so that each study and its associated objects can be inspected independently while preserving the relationships between objects:

```
dbgap/
├── studies/           one directory per dbGaP study, containing its objects
├── collections/       dbGaP collections (logical groupings of studies)
├── collections.json   aggregate summary of collections and their members
└── web/               supporting dbGaP reference pages (HTML/PDF)
```

At a glance, the Challenge Dataset covers:

Object	dbGaP Prefix	Approx. Count
Studies	phs	3,578
Phenotype datasets	pht	16,070
Variables	phv	465,853

Genotype datasets	phg	2,122
Documents	phd	8,079
Analyses	pha	17,277
Collections	phs (collection hosts)	23

The total size of the snapshot is approximately 6.3 GB unpacked (1.6 GB Compressed), dominated by approximately 466,000 variable JSON files and per-object HTML captures.

Task 1: Submission JSON Structure

To be considered for evaluation in this challenge, participants must upload a submission file containing responses for up to 1,000 dbGaP study variables comprising the evaluation set. Participants must also submit a description of their system, including its components, technical documentation, and deployment and execution instructions, in accordance with the System Submission requirements provided below.

Each response submission file should be a UTF-8-encoded JSONL file, with each line being a JSON object or record corresponding to a dbGaP study variable. Participants can submit up to three (3) submissions for the task. All submissions must conform to the required format described below and must be uploaded through the Challenge Portal.

Participants will be provided with the evaluation set of study variables on **January 18, 2027**. Participants should then develop variable mappings for all study variables in the evaluation set and submit their completed submission file(s) through the Challenge Portal before January 31, 2027.

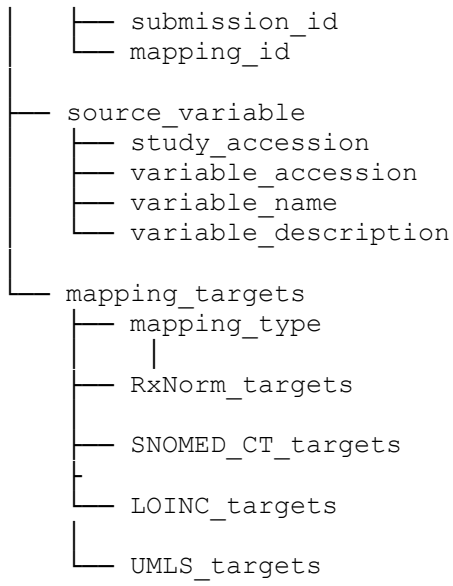
Each JSONL record represents the mapping of one dbGaP study variable to one or more concepts in the permitted target source vocabularies.

The structure consists of three primary components:

- Metadata
- Source Variable
- Mapping Targets

The Submission Record looks like this abstractly:

```
├── metadata
│   ├── team_id
```



1. Metadata

The *metadata* section identifies the participant and the specific mapping being submitted.

- **`team_id`** — Unique identifier for the participating team.
- **`submission_id`** — Identifies the particular system submission.
- **`mapping_id`** — Unique identifier for the variable-mapping task.

2. Source Variable

The source_variable section identifies the dbGaP study and variable being mapped and provides a basis for **augmenting the source information with related study and variable data from the Challenge Dataset**.

It includes:

- **`study_accession`** — dbGaP study identifier.
- **`variable_accession`** — dbGaP variable identifier.
- **`variable_name`** — Variable name.
- **`variable_description`** — Original dbGaP variable description.

Participants may augment this source information with relevant study metadata, variable metadata, data dictionary information, descriptions, value sets, and other contextual information available in the challenge dataset.

This establishes the **source information and study context** that the system can use to determine the variable's underlying semantic meaning and identify the most appropriate standardized source vocabulary concept(s).

3. Response

The response section contains the system's interpretation of the variable and its proposed ontology mapping.

Mapping Type

The **mapping_type** indicates the nature of the mapping, such as:

- `Exact` — The variable maps directly to an existing ontology concept.
- `Related_compound` — The variable represents multiple concepts that should be mapped separately.
- `Related_partial`— Only part of the variable's meaning can be represented by available ontology concepts.
- `None` — No adequate target concept exists; the system should return *ONT_NONE*.

Selected Targets

The **selected_targets** section identifies the ontology concept or concepts that the system determines provide the best representation of the dbGaP variable (or component variable).

For each target, the system provides:

- SNOMED CT: List of targets in the source vocabulary
  - **concept_id** — Identifier for the selected ontology concept.
  - **concept_name** — Name or preferred term associated with the concept.
- RxNorm: List of targets in the source vocabulary
  - **concept_id** — Identifier for the selected ontology concept.
  - **concept_name** — Name or preferred term associated with the concept.
- LOINC: List of targets in the source vocabulary
  - **concept_id** — Identifier for the selected ontology concept.
  - **concept_name** — Name or preferred term associated with the concept.
- UMLS: List of targets in the source vocabulary (if no exact match to SNOMED CT, RxNorm, or LOINC)
  - **concept_id** — Identifier for the selected ontology concept.
  - **concept_name** — Name or preferred term associated with the concept.

This allows the challenge to evaluate more than simple terminology matching. A high-performing system should demonstrate that it can:

- Understand the meaning of a raw dbGaP variable;
- Disambiguate abbreviations and contextual terminology;
- Identify the appropriate standardized vocabulary concept(s);
- Assess the degree of semantic correspondence;

System Submission

By January 15, 2027, participants must submit a Docker container image(s), a description of their system and its components, together with sufficient technical documentation and instructions to enable NIH to reproduce, deploy, and evaluate the system within an NIH-managed AWS environment. All software, models, indexes, data resources, documentation, and other materials that comprise or support the submitted system must have been created or last modified on or before January 15, 2027. A link to the participant's GitHub/Docker repository for the solution will be accepted.

Submissions should describe the system architecture and components, including retrieval, ranking, embeddings, indexes, models, LLM inference, and any other software or data resources required to execute the system. Participants may design their systems as a single containerized application or as multiple containerized services, including distributed architectures in which different components run on separate compute nodes. Participants should also provide deployment and execution instructions, along with an estimate of the computational and storage resources required (e.g., CPU, memory, GPU, and storage).

For evaluation, NIH will provision and manage the underlying AWS infrastructure, including compute, storage, networking, and orchestration. Participants will not be required to provision or manage AWS resources, Kubernetes clusters, or Terraform infrastructure.

To support verification of the January 15, 2027 submission deadline, participants may be required to provide cryptographic hashes or other verifiable records for software, models, indexes, data artifacts, and other system components identified in their submission. When requested during the evaluation phase, the corresponding executable or data artifacts must match the versions documented and verified at the submission deadline.

Model and Resource Restrictions: Submitted systems may use only publicly available, open-weight Large Language Models (LLMs), open-source software, and/or other publicly available computing resources. Systems may not invoke external APIs to proprietary frontier models (e.g., commercial hosted LLM endpoints) or rely on non-public, restricted-access software, models, datasets, or other resources during execution. Any pretrained/fine-tuned models used by the system must be included within the submitted Docker images.

- ⊄ Docker Image Execution and Network Restrictions

- The challenge organizers reserve the right to run the submitted Docker-based system and verify system reproducibility, resource utilization, and compliance with challenge rules. Specifically:
- External Calls: Submitted Docker Images will be executed in a completely isolated environment with all external network calls blocked. Limited access will be permitted to NLM-hosted services, such as the UMLS Terminology Services (UTS), and other challenge specific services. Details of these approved services and endpoints will be made available when the challenge is published. This zero-egress policy (excluding the permitted services) prevents both data leaks and runtime reliance on external computing resources, tools, or services.

Additional technical guidance, including submission formats, deployment specifications, resource limits, verification requirements, and evaluation procedures, will be provided to participants sufficiently in advance of the January 15, 2027 submission deadline.

Evaluation and Challenge Winner Selection

To determine the challenge winners for this task, participants will submit their developed systems and a submission file containing mappings for 1,000 dbGaP study variables comprising the evaluation set. Evaluation will be conducted in two phases.

Phase 1: System Selection

The evaluation dataset will be a subset of 200 study variables for the initial system-selection phase, which will identify the top-performing systems for more extensive evaluation in Phase 2.

All participant submissions will be evaluated against an expert-verified reference set of target source-vocabulary concepts. The primary evaluation metric will be Mapping Alignment Accuracy (MAA), which is defined as the Percentage of dbGaP study variables for which the system identifies the expert-verified target concept(s).

Each evaluated study variable receives one point when the system's predicted mapping exactly matches the expert-verified reference mapping (whether exact or partial, or ONT_NONE) and zero points otherwise. For a compound variable, where no single target concept adequately represents the variable's meaning, the variable may be decomposed into multiple distinct concepts, with each constituent concept mapped separately. In this case, the study variable is awarded one point only if all required constituent concepts are correctly identified and mapped to the expert-verified target concepts. If one or more of the component mappings is incorrect the overall mapping score will be zero.

A higher MAA score indicates better-performing systems.

The top N systems (expected to be five), based on Phase 1 performance in terms of MAA, will advance to Phase 2.

Phase 2: Extended Evaluation and System Deployment

In Phase 2, each of the top-performing (expected to be five) Docker-based solutions will be installed and configured in an NIH-provided AWS cloud environment to ensure a standardized evaluation environment. The systems will be evaluated using a private set of dbGaP study variables, as well as criteria related to usability, and operational performance.

Phase 2 will use the following evaluation criteria:

- MAA score: Measures the Percentage of dbGaP study variables for which the system identifies the expert-verified target concept(s)
- Ease of Use: Assesses the effort required to deploy, configure, operate, and use the system in the standardized NIH cloud environment.
- Compute Time: Measures the computational time required to generate responses.
- Compute Effort: Measures the computational effort required to generate responses, including the resources and processing required by the system.
- Memory Footprint: Measures the memory and other computational resources required to operate the system and generate responses.

MAA will serve as the primary metric for determining final system rankings and identifying the overall winning systems. In the event of a tie in this metric, the additional evaluation criteria listed above may be used to differentiate systems and determine the final ranking.